

7.1.18. Find exact values for each of the following quantities without using a calculator.

- a. $\log_3 81$
- b. $\log_2 1024$
- c. $\log_3(1/27)$
- d. $\log_2 1$
- e. $\log_{10}(1/10)$
- f. $\log_3 3$
- g. $\log_2(2^k)$

a. $\log_3 81$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\log_3 81 = y$$

$$3^y = 81$$

$$3^1 = 3$$

$$3^2 = 3 * 3 = 9$$

$$3^3 = 9 * 3 = 27$$

$$3^4 = 27 * 3 = 81$$

$$\text{Hence, } \log_3 81 = 4$$

Solution: $\log_3 81 = 4$

b. $\log_2 1024$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\log_2 1024 = y$$

$$2^y = 1024$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1024$$

$$\text{Hence, } \log_2 1024 = 10$$

Solution: $\log_2 1024 = 10$

c. $\log_3(1/27)$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\log_3(1/27) = y$$

$$3^y = 1/27$$

$$3^3 = 27$$

$$1/27 = 1/3^3 = 3^{-3}$$

$$\text{Hence, } \log_3(1/27) = -3$$

Solution: $\log_3(1/27) = -3$

d. $\log_2 1$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\log_2 1 = y$$

$$2^y = 1$$

$$2^0 = 1$$

$$\text{Hence, } \log_2 1 = 0$$

Solution: $\log_2 1 = 0$

e. $\log_{10}(1/10)$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\log_{10}(1/10) = y$$

$$10^y = 1/10$$

$$1/10 = 10^{-1}$$

$$\text{Hence, } \log_{10}(1/10) = -1$$

Solution: $\log_{10}(1/10) = -1$

f. $\log_3 3$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\log_3 3 = y$$

$$3^y = 3$$

$$3^1 = 3$$

$$\text{Hence, } \log_3 3 = 1$$

Solution: $\log_3 3 = 1$

g. $\log_2(2^k)$

Definition of Logarithms and Logarithmic Functions:

$$\log_a x = y \leftrightarrow a^y = x$$

$$\begin{aligned}\log_2(2^k) &= y \\ 2^y &= 2^k\end{aligned}$$

$$\text{Hence, } \log_2(2^k) = k$$

Solution: $\log_2(2^k) = k$